Technical PDF to LaTeX

A faithful, resumable conversion framework for the dual DGX Spark

1. Recommendation and agreed scope

Build a small Python application around a specialist document converter, a persistent source record and bounded model review. Reuse the Quant Wiki source-backbone contracts and implementation prompts; adapt their output to faithful editable LaTeX. Use the selected local GLM-5.3-Flash endpoint for reconciliation and optional OpenRouter vision for difficult material.

Content and mathematical fidelity take priority; layout matching is excluded. Figures may remain extracted images. Papers and complete books are the main workload; scanned and mixed PDFs are included. You have authorised all source content in the optional cloud mode. This is the revised design and implementation plan, not an installed or tested application.

Decision	Starting point
Converter	Probe MinerU Basic first, then Standard on identical pages. Standard is the initial quality candidate. Test native LaTeX export before building a replacement renderer.
Reusable core	Immutable artifacts, conversion manifests, source spans, stage attempts, durable state and bounded book jobs from walkthrough 1.1 and the 5 October skill.
Local model	GLM proposes source-linked transcription and structural changes. Its actual served identity, request limits and vision support require qualification.
Dual-Spark operation	One coordinator and registry writer; separate heavy conversion and GLM phases initially. Measure per-node headroom before allowing coexistence.
Optional cloud	Local-only, flagged-material or cloud-first mode per document, within a configured budget. No repeated content-consent prompts after enabling the mode.
Output	Compilable LaTeX project, extracted figures, reconstructed PDF, source map and an explicit coverage, editability and review report.

What revision 1.1 changes

The newly attached standalone walkthrough is revision 1.1 and matches the canonical source recorded by **quant-wiki-spark-pilot**. The attached ZIP is older revision 1.0. Sections 3 and 3A document precedence and the reuse map; sections 6 and 11-16 turn that reuse into records, deployment checks and milestone gates.

MinerU already exposes a LaTeX renderer, so the first implementation should measure and wrap it where suitable. [1-3] The main new work is complete source accounting, traceable mathematical review and reproducible export. Original author macros cannot generally be recovered uniquely; the target is equivalent editable typesetting.

Next deliverable after review: an adapted probe package, followed by the source backbone and one-paper LaTeX prototype. No model training, whole-book generation request or wiki migration is required.

2. What fidelity means in this framework

The authority is the supplied edition of the PDF. Faithful transcription and mathematical correctness are separate: if the source contains a questionable sign, the transcription preserves it and records a possible source error separately. No automatic simplification, notation modernisation, proof completion or silent correction is permitted.

Content	Required behaviour
Prose and inline mathematics	Preserve wording, symbols, emphasis with mathematical meaning, footnote attachment and reading order. Fix line-wrap artefacts only with evidence.
Displayed mathematics	Editable LaTeX, including limits, grouping, matrices, cases, multiline continuations and original visible equation labels.
Tables and algorithms	Editable structure where feasible; retain headers, units, notes, indentation and row/column associations. Image fallbacks are unresolved transcription work.
Figures	Extract the original asset when complete; otherwise crop the full visual region, including labels, legends and panels. Captions remain editable text.
References and book matter	Retain citations, bibliography, theorem/proof structure, appendices, front matter and back matter. Preserve original index page references.
Layout	A clean article/book template. No target for original fonts, margins, columns, float positions or page breaks.

Four independent measures of completion

Coverage: every physical page and inventoried content object has a disposition. **Editability:** text, mathematics and tables are distinguished from image fallbacks. **Build:** the exported project compiles with its recorded environment. **Review:** exact checked objects/pages and remaining issues are reported. None of these measures substitutes for the others.

Accounted-for pages include explicitly unresolved pages; this is distinct from recognised, exported and verified pages. A draft can compile with marked fallbacks or unresolved symbols and must not be labelled a completed editable reconstruction. A reviewed subset identifies exactly what was checked. A reviewed document names its protocol and exceptions; it is not a guarantee of error-free transcription.

Source-first acceptance

Every substantive object is either included as editable content, included as an allowed figure image, explicitly excluded as a layout artefact, or unresolved. Running headers and footers remain in the evidence record even when omitted from the reading edition. No repeated text is discarded solely because it repeats.

When original TeX is legitimately available from an author or repository, an optional source-import path should check that it matches this PDF version before reuse. A preprint with different equations is a separate source, not a substitute for the selected edition. Source lookup is an optional optimisation, not a prerequisite for the local converter.

3. Recovered material and precedence

There is reusable project material. The attachments now provide direct copies of the Quant Wiki walkthrough, prompts and templates. They strengthen the source-recording design already recovered through the earlier project contracts. They do not contain a finished converter application.

Input inspected	Finding and use
Standalone walkthrough (2).md	**Revision 1.1, 25 September 2026.** Explicitly supersedes 1.0. Its SHA-256 matches the canonical walkthrough recorded in the installed skill. Use its body, five embedded prompts and two templates.
Quant-Wiki-Spark-Pilot-Guide.zip	**Revision 1.0, 24 September 2026.** Eight Markdown files: walkthrough, START-HERE, four prompts and two templates. Retain as historical reference; do not execute its superseded combined digest/retrieval prompt.
quant-wiki-spark-pilot	**Skill revision 1.0, 5 October 2026.** Preserves walkthrough 1.1 plus applicable Build Contract 2.2 state/evidence rules. Use its five current prompt assets and source-record/evaluation reference.
run-two-spark-models	**Research snapshot, 4 October 2026.** Use deployment qualification, model/runtime identity, per-node resource checks and preservation of working services. Historical recipes are not proof of a live deployment.

Which requirements control this project

Your current instructions control scope: faithful LaTeX, GLM and two Sparks, with optional unrestricted-content cloud assistance. Apply the walkthrough 1.1 evidence and recovery rules within that scope. The old one-Spark/Qwen baseline remains historical; it does not require changing your model back. Existing verified project changes must be compared before copying any prompt or template.

The old digest rules permitting condensed prose or proof outlines do not apply to the LaTeX reconstruction. Preserve complete source content. Likewise, the older ZIP's autonomous proof/error-audit suggestions are superseded: source mathematics must not be silently corrected.

What remains unavailable or unverified

The ZIP supplies no application source, implemented schema, converter adapter, dependency lock, registry, probe results or executed RUN-PILOT.md. These attachments contain no Open WebUI Tool implementation. Reuse of that project remains conceptual until its code is supplied or located. No Spark installation, conversion benchmark or endpoint probe has run for this plan update.

Input fingerprints (full-byte SHA-256):

```
Standalone 1.1
727a0195ae2269db38057fbb9ba29e5fab6962c962930539235d36604b8cb844
Attached ZIP 1.0
d031ba2c764b31e41e2c677768fe01e3f42893e4668bd5fc54a340fa62e63c2f
```

3A. Reuse map for implementation

Use the current assets from the 5 October skill, which match the standalone 1.1 walkthrough. Stage reviewed, adapted copies in the new project; preserve the predecessor text and an adaptation log. The entries below are planned reuse, not claims that code already exists.

Current asset	Adaptation for pdf2tex-workbench
01-inventory-and-probe.md	Reuse read-only inventory, endpoint checks, isolated converter probes and measured paper/chapter evaluation. Substitute the actual GLM/two-node profile. Respect the selected cloud mode; remove the obsolete blanket cloud exclusion.
02-source-backbone.md	Reuse immutable registration, five-record contract, real page locators, raw outputs, locking, restart recovery and backups. Add typed page/block content and LaTeX export manifests.
03-source-retrieval.md	Reuse structural boundaries and dependency spans for bounded conversion/review requests. FTS5, embeddings and question-answering are deferred to an optional wiki bridge; they do not block LaTeX.
04-technical-digests.md	Reuse evidence checks, raw-versus-normalised provenance, separate verification status and rejection of stale spans. Replace summarisation with complete transcription; no shortened proofs or omitted exposition.
05-books-graph-and-release.md	Reuse bounded full-book jobs, gaps/overlaps, front/back matter, portable assets and reviewed versioned releases. Export a LaTeX package. Graphify and Obsidian release work remain optional and downstream.
Technical-Digest.md	Extract the notation/assumptions, algorithm controls, numerical settings and evidence checklist into a Fidelity-Review template. Replace digest treatment with reconstruction disposition; retain exact review scope.
Pilot-Questions.md	Turn source-verified questions into transcription assertions: appendix steps, theorem conditions, table notes, scoped symbols and cross-page dependencies. Preserve complete expected evidence and explicit unconverted/out-of-scope status.

Reused contracts, new components

Retain Artifact, Conversion, SourceSpan, Chunk and StageAttempt semantics, original hashes, edition metadata and known legacy IDs. Implement or adapt the source core after inspecting actual project code. Build the converter adapter, typed LaTeX content, renderer wrapper, review-decision path and cloud adapter where no tested implementation exists.

Keep **pdf2tex-workbench** separate from the existing Quant Papers Vault and benchmark environments. A later bridge exports complete source Markdown, assets and source IDs through a reviewed ID/path crosswalk. Never inherit old review approval for a new conversion generation.

What this reuse removes from the critical path

The source identity, evidence, recovery and staged-development contracts no longer need to be designed from scratch. Embeddings, a graph, an ontology and digests are unnecessary for faithful LaTeX. If the wiki bridge later adds them, preserve its order: full-source retrieval must work before digests, and graph features remain optional.

4. Conversion tools and selection rules

Candidate	Role and evidence-based limits
MinerU 4.0.12	Primary candidate. Basic uses smaller OCR/formula/table models; Standard adds a vision-language model; Advanced spends more inference effort. Flash is unsuitable as the final editable-math path: display equations can remain images. [1-4]
Marker / Surya	First independent challenger for difficult pages and inline mathematics. Current balanced mode handles inline math; disabling OCR also skips equations and scans. Native outputs include structured JSON, not a promised complete TeX project. [7]
Docling	Structured alternative with formula-to-LaTeX enrichment. ARM64 is documented; exact accelerator/dependency combinations still require testing. Keep it as a fallback, not a second mandatory full-book pass. [8-10]
Chandra OCR 2	Optional page-level challenger for difficult scans. Its project supports page ranges and image outputs. Exact Spark memory and runtime compatibility remain unqualified. [11]
pix2tex / LaTeX-OCR	A possible recogniser for individual equation crops; not a replacement for document reading order, prose, tables or references. [12]
Nougat	Optional historical comparison, not the default. Its documentation records missing-page heuristic problems; a skipped page is unacceptable without explicit accounting. [13]

This is a selection order, not an accuracy ranking on your books. Published benchmarks and vendor claims do not identify the best option for dense financial mathematics on the Sparks. The pilot decides whether the simpler Basic path is sufficient, whether Standard reduces correction effort, and whether the challenger finds important errors missed by both.

Keep the first comparison small

Run Basic and Standard on exactly the same representative pages. Compare the native MinerU LaTeX export with its structured output, including inline formulas, equation labels and assets. Introduce Marker only on that bounded sample or on observed weak cases. Use Advanced for unresolved cases if it demonstrates improvement. Do not install every candidate or run every page through every model.

PyMuPDF provides independent native text, character/block geometry, rendering and asset evidence; it is not by itself a math-to-LaTeX recogniser. Retaining a second view helps detect extraction omissions and bad font mappings. [14]

Version and licence record

Record exact package versions, model revisions, hashes, runtime images and code/model licence files at installation. Code and model-weight licences can differ; a personal prototype must not silently be treated as an approved bank deployment. The initial package should reuse installed software where suitable and download only the selected models.

5. The processing pipeline

Stage	Work and durable output
1. Register	Copy the selected PDF, hash it, count physical pages, record metadata/page labels, and identify encryption or corruption. Preserve the input unchanged.
2. Inventory	Examine page text and images; classify each page as native, scanned, mixed or uncertain. Record rotations, columns and candidate regions.
3. Recognise	Run the qualified converter in bounded windows. Save raw structured output, text, crops, figures, requested/observed page sets and engine diagnostics.
4. Reconcile	Map results to project blocks, join true continuations and preserve alternative candidates. GLM proposes bounded changes and cross-reference links.
5. Check and review	Check coverage, formula syntax, labels, references and differences. Use a second recogniser or cloud vision on uncertain material; present unresolved cases to the user.
6. Assemble	Render a stable LaTeX project from the chosen block versions. Compile chapters and the whole document in a fixed environment.
7. Release	Export source, PDF, assets, source map and a truthful quality report as a checksummed version. A later rerun produces a new version.

The input path depends on each page

For a born-digital page, retain native characters and geometry alongside recognition results. Selectable text can still have broken glyph mappings. For scans, render suitable image resolution and perform recognition. For mixed pages, preserve trustworthy native text and recognise image-only regions. An empty extracted-text result is a routing signal, not permission to omit the page.

Begin with 300 dpi rendering for mathematical scans/crops as an engineering starting point, then increase or tile only when measured failures justify it. Save pixel dimensions, crop coordinates and all transforms. Deskew or contrast variants remain derivatives of the original, never replacements. Avoid aggressive cleanup that can erase dots, primes and minus signs.

Resource and output boundaries

Initial proposed conversion windows contain about 8-16 pages, adjusted after memory measurements. They carry adjacent-page context for split paragraphs or equations, with one designated owner for each output object. Overlap must not duplicate content. Complex oversized blocks are split only at verified structural boundaries and retain continuation links.

Store assets and structured content together: saving only a JSON representation is not enough when image bytes or references live outside it. The adapter validates actual paths and hashes rather than guessing asset filenames. [3]

6. Document records and LaTeX assembly

Retain the five Quant Wiki record contracts. Use a small typed representation between raw extraction and rendering; SQLite owns state while immutable JSON and asset generations hold substantial content. Preserve the converter format unchanged beside the project representation.

Record / extension	Contract and purpose
Artifact / Conversion	Stable source identity, full-byte hash, edition, original filename and page count; converter/model/runtime/configuration, requested and observed coverage, assets and checksummed manifest.
SourceSpan	Artifact and conversion IDs; physical pages, printed labels separately, real block IDs or declared text offsets and hashes; optional source boxes/crops with explicit transforms.
Chunk	Retain the compatibility contract: ordered primary spans, separate context spans, section path, grouping version and token budget. Populate for bounded review when needed; no vector index is required.
StageAttempt	Stage, input-generation fingerprint, queued/running/succeeded/failed/interrupted/blocked state, actual settings, times, retries, errors, resources and output manifest. Do not create a second incompatible Attempt identity.
Page / Block; Candidate / Revision	Add typed content, reading order, continuations, raw/chosen text or mathematics; alternatives, evidence, base hash, decision, reviewer and superseded version.
ReviewIssue / Export	Add uncertainty and resolution records, source-linked acceptance decisions and a versioned TeX build/quality manifest. Introduce schema changes through explicit migrations.

Use zero-based physical pages internally and one-based PDF pages in the interface. Printed labels are independent strings, including Roman numerals. Store boxes in PDF points in a declared unrotated page frame and retain raster/rotation transforms. Declare text-offset conventions. Test every adapter with a known rotated fixture.

Preserve structure; render it deterministically

Paragraphs contain ordered text and inline-math runs. Displays retain lines, visible tags and all source boxes. Tables retain cells, spans, headers and notes; captions and footnotes have attachment links. The renderer escapes prose and emits mathematics through controlled templates.

Use article/book templates, chapter files and a small shared preamble. Preserve source-visible equation/theorem/figure/table labels separately from generated counters. Internal labels are source-scoped and unique; visible labels may repeat. Unresolved references stay literal and flagged. Bibliography transcription comes first; optional BibTeX enrichment must not invent metadata.

Keep original index page references, linking original-page anchors where possible. A new index for reconstructed pagination is separate. Test the upstream LaTeX renderer first; extend only failures of the fidelity contract, including asset packaging, stable labels and chapter assembly.

7. Mathematical verification

Compilation is necessary but weak evidence of transcription accuracy. A missing minus sign, an exchanged subscript, or a changed integration bound can compile perfectly. A model recognising a familiar formula can also replace an unusual source expression with a more plausible one. The checks must keep the visible source authoritative.

Check	What it catches; what it does not prove
Coverage and region comparison	Missing or duplicated paragraphs, equations and notes. Review complete pages too, because a crop-only system cannot see an undetected object.
Text/formula candidate disagreement	Symbol changes, missing terms, different matrix sizes or limits. Agreement can still be a shared mistake.
Syntax and structure	Unbalanced delimiters, broken environments, truncated matrices/cases, incomplete outputs. Valid syntax does not prove the right expression.
Source crop versus rendered formula	A focused visual check of the chosen LaTeX. Global pixel similarity is not an acceptance rule because fonts differ and tiny signs matter.
Context and notation	Inconsistent symbol definitions, numbering or suspected source mistakes. These create review issues; they do not authorise semantic corrections.

High-priority error classes

Check signs; factors; differentials; integral and sum bounds; fraction scope; subscripts and superscripts; accents and primes; bold/calligraphic/blackboard-bold symbols; conditional-expectation bars; set-builder separators; matrix rows; cases conditions; and multiline continuations. Distinguish visually similar characters such as 1/l/I, 0/O, v/nu and p/rho. Inline mathematics must have its own detection and accuracy measures.

How an uncertain expression is resolved

Save the original crop with a small amount of neighbouring context. Obtain a second transcription without showing the first candidate where an independent reading is useful. Then compare candidates and their rendered forms. If disagreement remains, enlarge the crop, inspect the whole page, or ask for human review. At most two targeted re-recognition attempts per issue are an initial policy, not an upstream limit.

Do not use an uncalibrated model confidence percentage as an automatic acceptance threshold. Prioritise issues from observable evidence: candidate disagreement, missing region, truncation, syntax error, suspicious symbol change or source-reference mismatch. Correlated agreement between two calls to the same model is a triage signal only.

For an unattended draft, uncertain mathematics remains marked and an original crop can make the document readable. It is counted as untranscribed until resolved. Suggested mathematical corrections belong in a separate errata/annotation file, never silently in the faithful edition.

8. Full books, scans and long context

A book is a durable collection of page and chapter jobs. Its length should affect the number of jobs and disk usage, not require a single enormous prompt. The global structure holds the edition, chapter order, page-label map, reference index and notation references. Individual jobs receive only the source evidence needed for their bounded task.

Chapter boundaries and cross-page content

Build a provisional chapter map from bookmarks, contents pages and heading evidence; record uncertainty rather than treating the contents as infallible. Keep physical-page windows for conversion and semantic boundaries for reconciliation. A paragraph, proof, table or equation crossing a window boundary is represented by linked source spans and assembled exactly once.

When a table continues, preserve repeated headers in the evidence and reconcile the body rows without duplicating them. Footnotes remain tied to their markers. Repeated equation labels in different chapters get distinct internal IDs. Theorems and exercises keep their original numbering and text; no global notation substitution is applied simply because a symbol has a different meaning elsewhere.

Long-context use

Use GLM on paragraphs, connected equations or short sections. Start with a conservative request envelope, for example no more than about 16,000 input tokens plus a separately reserved output/reasoning budget, then tune from real usage. This is a proposed task size, not a claim about your maximum served context. Count with the deployed tokenizer where possible and account for images separately.

A chapter-level symbol map records exact source definitions and IDs; it is not a replacement summary of the chapter. When definitions are uncertain, pass the actual source spans. A final cross-reference pass uses compact structural records rather than resending the whole book. A 1M advertised context window would not by itself solve output limits, omissions or reviewability.

Coverage that survives scale

The status report lists each chapter, requested and accounted pages, editable/unresolved objects, source artefacts intentionally excluded, and the next action. Front matter, appendices, bibliography and index are first-class jobs. Blank pages are recorded as blank. A selected chapter must be labelled an extract, with out-of-scope references retained.

For scans, track page-specific quality: skew, rotation, faint text, bleed-through and low resolution. Use a known-TeX rasterised fixture to test mechanisms, plus a genuine difficult scan for realistic error rates. A rasterised digital page is only a controlled proxy for a scanned book. Mixed documents must be routed page by page.

Cache page renders on disk with a bounded policy; do not hold hundreds of full-resolution pages in memory. Preserve originals, chosen crops and review evidence so a completed run remains auditable after temporary images are removed.

9. GLM requests and the review workflow

Keep orchestration in ordinary Python. The model receives bounded tasks and returns proposals; it does not manage directories, run the compiler, choose shell commands or decide that a book is complete. This reduces the chance that a long agent conversation silently changes the conversion rules.

Capability qualification for your actual endpoint

Use your configured OpenAI-compatible address, currently reported as http://192.168.1.100:8888/v1. Discover the served model ID instead of assuming an older alias. Test text generation, output truncation, structured responses and actual request budgets. Save the serving profile and checkpoint identity when exposed.

Z.ai documents native image input for GLM-5.3-Flash. That does not establish visual support in the Mia-AiLab EXL3 checkpoint and patched TensorFold path. A vision probe must distinguish images containing different, unprompted symbols/numbers, including a blank control; HTTP success is insufficient. Until it passes, use GLM for text/structure and the converter or a qualified cloud endpoint for visual transcription. [15]

A proposed correction contract

```
{
  "block_id": "bookA:conv03:p0128:eq004",
  "base_revision_hash": "<hash of current content>",
  "operation": "propose_replacement",
  "latex": "<source-faithful candidate>",
  "visible_label": "4.17",
  "evidence_crop_ids": ["crop-128-04"],
  "uncertain_spans": [],
  "issue_codes": [],
  "short_note": "<only if needed>"
}
```

Validate every required ID, hash and field; reject missing/duplicate IDs, invented blocks and truncated outputs. This establishes a valid proposal, not its fidelity. Meaning-changing replacements require visual source evidence and a recorded acceptance decision; a text-only GLM cannot settle an unseen glyph. Unresolved discrepancies stay provisional. Preserve retries and repairs separately, and reject edits against a stale revision hash.

Review interface

Begin with an HTML review pack showing page overlays, original crops, LaTeX, compiled expressions and issue reasons, linked to whole pages. Milestone 1 also includes a small CLI or structured-file import for accept/edit/defer decisions against block revision hashes. A convenient browser editor comes later; the first prototype can already close the review loop.

Corrections are versioned overrides against block hashes and are reapplied only when their source still matches. Exported TeX can be edited freely, but such edits form an external fork unless deliberately imported. Regeneration never silently overwrites a user-edited export. Group repeated font-mapping problems to make review efficient while retaining affected locations.

10. Optional OpenRouter and WebUI integration

Your permission covers all document content, including full pages, crops and text. The proposed cloud feature therefore needs no redaction stage. It remains optional because cost, availability and local operation are useful choices. Set the mode per document: **local-only**, **flagged-material** or **cloud-first**. Once enabled, calls proceed within that job's selected limits without per-equation prompts.

Send explicit evidence

Prefer locally rendered page/crop images, adjacent source text and stable IDs. OpenRouter supports image content in chat-completion requests. Whole-PDF input can invoke an intermediate parser; the documented OCR route forwards at most eight extracted images and can strip them for text-only models. Therefore a book upload does not prove that every formula image reached the model. If PDF input is used, choose and record its engine explicitly. [16-17]

Choose models through a small measured comparison

At setup, snapshot available model/provider IDs, image support, context/output limits, structured-output support and current prices. Qualify two strong image-capable candidates on the same small source-verified set, with candidate-blind transcription where appropriate. Select on formula fidelity, unflagged errors, latency and total review cost. Do not hardcode a supposedly best cloud model before this test. [18-19]

During comparison, use one explicit provider per model and disable fallbacks. Provider preference order alone is not an allow-list. For production, optional fallback is restricted to prequalified profiles and recorded per attempt. Parameters must be supported by the actual endpoint; model metadata alone is insufficient. [20]

Spending and failed calls

Require an explicit document ceiling before enabling paid calls; leave it unset in the initial configuration. Track images, input/output/reasoning tokens, parser charges, retries and failed/uncertain attempts. Reserve a conservative amount before dispatch and reconcile returned usage. An API timeout is not proof of no charge: retain an outcome-unknown record rather than promising exactly-once remote execution. Unit-price filters are not total-budget controls. [20-21]

Pause cloud scheduling at the ceiling while preserving progress and allowing eligible local work. Bound retries and concurrency; start with one in-flight call. Use a provider-side budget as a second boundary when available. No cloud call is needed to compile or inspect a completed export.

Open WebUI 0.12.0 as an optional front end

A thin tool submits a durable job and returns its ID; status, resume and download are short operations. The worker survives a closed browser. Current Tools/Valves documentation is useful, but the exact 0.12.0 file and event interfaces still require qualification. Keep API keys in service configuration, not prompts. Reuse the earlier consent concept, with a document scope distinct from a general chat scope. [22-23]

11. Deployment on your two Sparks

The target remains your reported dual-Spark GLM setup. Earlier hardware, address and TensorFold details are configuration leads to verify, not results of this task. The two-Spark skill supplies a qualification procedure; its 4 October snapshot does not override newer user-reported equipment or prove that a particular service is running.

Profile	Initial operating policy
Local extraction	Use a measured idle window. If needed, stop only the identified GLM service through its verified procedure; run one qualified converter, persist outputs, then release its model memory. No automatic broad process termination.
GLM reconciliation	Restore the existing recorded two-node profile. Start with one bounded request and ordinary context. No heavyweight OCR model competes until coexistence is measured.
Cloud continuity / CPU path	A lightweight coordinator may process saved images through OpenRouter or run qualified CPU extraction with measured headroom. CPU allocations also consume unified memory.
Future parallel conversion	Only after qualification, use otherwise idle nodes for separate windows. Return checksummed outputs to one coordinator and one authoritative registry writer. No distributed task service is initially needed.

Apply run-two-spark-models to qualification

Inventory both actual nodes, active jobs, model caches, OS/driver and usable memory. Preserve working launchers. Record exact checkpoint/quantisation, runtime commit and patches, image digest, cache and draft settings, request limits and concurrency. Discover the served model ID, then exercise a genuine generation. Test image understanding separately; HTTP success or model-name metadata is insufficient.

Two nodes do not create a transparent 256 GB GPU. Measure each node's MemAvailable plus process/framework peaks; weights, caches, workspaces and the OS share capacity. Aggregate nvidia-smi memory may be unavailable on Spark. [6] A checkpoint's disk size is not a runtime-memory estimate.

Separate prompt processing, time to first token, per-request decode speed and aggregate throughput. Record prompt/output sizes and concurrency. Per-request context includes input, reasoning and output; configured aggregate cache is not the request budget. Whole-book processing uses bounded jobs and does not require a 262K or longer-context deployment experiment.

Resource admission and recovery

Keep SQLite on one node's local SSD, never a network share or two writable replicas. Admit a work unit only when its measured peak plus reserve fits on every involved node. Pause new work on sustained pressure; use measured recovery conditions. An interrupted unit remains incomplete and resumes from durable state. Do not invent universal thermal limits.

Service stop/start hooks stay unset until exact working commands and rollback are verified. Preserve benchmarks and unrelated jobs. If the existing GLM service works, no cluster rebuild, model download, firmware change or serving-runtime migration is part of this project.

12. Installation and qualification plan

Keep the lightweight controller, converter and TeX builder isolated from the working model environment. A virtual environment isolates dependencies; a container supplies a separate userspace. Neither establishes ARM64/GB10 GPU compatibility. Reuse suitable existing installations after inspection.

Step	Action and exit condition
A. Inventory	Read both nodes' real configuration and active jobs; locate any existing QuantWiki-Pilot code, registry and RUN-PILOT.md at user-supplied paths. Distinguish planned, implemented, exercised and accepted work.
B. Controller	Use a supported existing interpreter or separately managed user interpreter. Install typed-record, HTTP and PDF-inspection dependencies into a new environment; freeze successful versions. No system-Python changes.
C. Basic probe	Start with the selected isolated MinerU Basic candidate. Verify actual backend/device and one paper plus a recorded book chapter. Successful CPU execution establishes neither Standard nor GPU support.
D. Standard comparison	Qualify the pinned ARM64 runtime on the same pages, recording model assets, runtime, accelerator, loading time, throughput and corrections. Use adequate measured headroom.
E. TeX builder	Fixed TeX Live/XeLaTeX, latexmk, fonts and controlled packages. Compile without network, API secrets or shell escape; restrict file access and resources. Reject unexpected inclusion/execution.
F. Adapters	Qualify actual GLM text/structured output and image capability independently. Add cloud capability/budget checks when enabled; WebUI follows a working durable core.

Worker and administrator responsibilities

Reuse the Quant Wiki worker boundary: an ordinary account without sudo or Docker-socket access, selected source copies, and approved releases outside its write scope. The administrator or authorised service account may prepare the isolated GPU runtime. Expose a narrowly scoped job/service interface or use an owner-launched bounded probe; do not grant the conversion worker a Docker socket to make containers convenient.

Check endpoint access in the real worker context and, if an agent sandbox is used, in that context too. Discovery and generation are distinct checks. A denied call is not permission to bypass the sandbox. A suitable existing account/setup need not be recreated.

Version and runtime qualification

The original design identified MinerU 4.0.12 as the candidate to qualify from primary documentation retrieved on 17 October. Its documented AArch64/Blackwell path and Docling's ARM64 support are compatibility leads, not successful local tests. [1-5,9-10] Inspect installed help and execution mode; pin exact packages, model revisions and image digests after a successful probe.

Check all requested pages and final-page output regardless of CLI defaults. LaTeX rendering is an SDK/export operation, not an assumed CLI flag. A satisfactory Basic path may proceed if Standard or local vision is unsupported. Preserve assets and raw JSON. RUN-PILOT.md records only commands exercised on that installation; unrun owner actions remain separate.

13. Pilot and acceptance criteria

Retain the selected first paper, The Smile Calibration Problem Solved, and a difficult chapter from Boyarchenko and Levendorskii's Non-Gaussian Merton-Black-Scholes Theory. The earlier design recorded 17 and 421 physical pages respectively; these were source-inspection observations, not conversion results. Confirm the same file hashes when staging the pilot.

Start with one paper and one complete, recorded chapter, then a whole book after the gates pass. Add a genuine scan when available plus a known-TeX fixture containing inline math, signs, accents, matrices/cases, a table with notes, columns, rotation and page-spanning content. A degraded fixture tests routing but does not represent every real scan.

Evaluation	Report with an explicit denominator
Coverage / editability	Expected, accounted, omitted and duplicated pages/objects; editable formulas and tables; image fallbacks separately. Accounted does not imply exported or verified.
Mathematics / prose	Inline/display detection separately; fully correct formulas on source-checked examples; prose errors, labels, reading order, table cells, assumptions and algorithm controls.
Uncertainty	Serious errors flagged and serious errors missed. Review scope is explicit; low review volume alone is not success.
Practical cost	Runtime by stage, download/load time separately where available, per-node peak memory, disk, API costs including failures, and human correction time.

Adapt the inherited evidence-question template

Write ten source-verified transcription checks before evaluating output. Include an appendix algorithm's initialisation/stopping rule, a numerical table with its configuration and units, theorem conditions, a cross-page equation condition, and a book definition used later. Record artifact/edition, physical pages, complete evidence and expected content; do not define gold by mutable chunk IDs.

Check that the reconstructed content preserves every required element and resolves to the correct source. Missing source data, unconverted scope and incorrect transcription are distinct outcomes. No question-answering subsystem is required. The wiki's 30-question retrieval suite belongs to a later retrieval integration and is not a LaTeX completion certificate.

Prototype and quality gates

Prototype: every selected page is accounted for, source links/assets resolve, exports compile, deliberate omissions and truncation prevent success, human overrides survive resume, and audited critical errors are corrected or explicitly unresolved. The source-backbone gate separately proves locking, crash recovery and backup/restore.

Quality: compare Basic and Standard on identical source-checked examples and then holdouts. Permit equivalent TeX formatting only when it preserves symbol distinctions, boldness, accents, grouping and indices. Select on measured correction burden; do not infer general accuracy from the small set. Whole-book operation initially tests completeness and restart behaviour with explicitly sampled fidelity; a stricter all-formula review remains selectable.

14. Durable state and failure handling

Adopt walkthrough 1.1 / Build Contract 2.2: local SQLite with checked WAL, foreign keys and **synchronous=FULL**. Hold **fcntl.flock** on a stable lock file for the mutating worker lifetime; never unlink/recreate it while held. Route online review/edit/pause mutations through the owning coordinator. Offline commands acquire the same lock or fail promptly; status readers remain independent. Keep transactions short, never across inference.

Write and resume protocol

Record an attempt, then write a unique staging generation on the destination filesystem. Close, flush, checksum and validate outputs with a self-describing attempt/input manifest; sync files and parent directories as supported. Atomically rename to an immutable generation, then commit its manifest and success in SQLite. On restart, reconcile interrupted attempts and renamed-but-uncommitted outputs by identity and hashes; quarantine invalid generations. File existence alone is not success.

Fingerprint input/code/configuration, requested page/span membership, preprocessing, model weights/revision and runtime, prompt/schema, grouping/chunk versions, generation/output settings and retry/repair policy. Include renderer dependencies for exports. A served alias is not immutable identity. Save raw requests/responses before cleanup and retain rejected candidates. Record provider provenance even when exact hosted replay is unavailable.

Failure	Required behaviour
Parser failure / missing page	Preserve completed windows, mark the affected unit incomplete and use bounded retries or a qualified alternative. Never silently accept a plausible prefix.
Resource pressure	Pause admission; retain completed state. An interrupted inference unit remains incomplete and is retried under the same accounting policy.
Cloud timeout / ceiling	Record unknown-charge outcomes, pause eligible calls and continue allowed local work. Do not promise exactly-once remote execution.
Compile failure / changed source	Retain logs and localise by block/chapter. Source changes create a new generation; meaningful repairs need source evidence and acceptance.
Human edits / stale evidence	Preserve prior accepted content. Reject stale hashes, wrong-source spans and unsafe asset paths; never overwrite user-edited exports silently.

Required recovery and fidelity tests

Interrupt during staging and between rename/commit; reject a concurrent writer; restore a backup; detect changed inputs, missing assets, duplicated overlaps, stale edits and invalid/truncated responses. Verify zero cross-model/cache-generation reuse and raw-versus-cleaned response accounting. Inject a missing minus, dropped equation and lost theorem condition; report validator sensitivity separately from recognition quality.

For backup, pause mutations while capturing a consistent SQLite backup plus the manifest of referenced immutable originals, conversions and records. Copy those files too; never copy a live main database alone while ignoring WAL. Restore separately and check hashes/links before release. Generated STATUS.md reports state; RUN-PILOT.md lists actually exercised commands, and proposed/unrun actions remain labelled. Test SSH-disconnect survival separately from reboot/crash recovery.

15. User workflow and deliverables

The first usable interface should be a command-line tool plus an HTML review pack. A local browser interface and Open WebUI connector can be added after the conversion core passes its pilot. All interface names below are proposed application commands, not installed software.

Operation	Behaviour
inspect / register	Show document type, page count and estimated work; register an immutable input and selected scope.
convert / resume	Run the selected profile, stream progress and resume compatible unfinished work.
status / pause	Report each stage, missing work, resource waits, costs and review counts; pause at a safe boundary.
review / export	Open source-linked issues and publish a versioned LaTeX project, draft PDF and quality report.

What a completed conversion package contains

Path within the export	Purpose
main.tex, preamble.tex, chapters/	Editable source, controlled macros and chapter files.
figures/, assets/	Extracted figures and any explicitly reported fallback crops.
reconstruction.pdf	The compiled reading edition, with its draft/review status.
source-map.json, manifest.json	Stable source IDs, page/block mappings, hashes and dependency versions.
quality-report.html, issues.json	Coverage, review status, known uncertainties and a navigable source comparison.
build instructions and logs	Exact environment/profile and successful compilation record.
optional source.md	Complete source text with math for later Quant Wiki ingestion.

The original PDF stays in the local source store; include it in a portable bundle when a self-contained review copy is wanted. Source links must resolve within that bundle or be clearly labelled external. A package with cropped equations must distinguish figure images from unresolved mathematical images.

Proposed configuration, in plain language

Select the input and page scope; converter profile; local GLM endpoint; optional cloud profile and spend ceiling; maximum work-unit size; resource-pause policy; and review protocol. Keep local-only as the ready-to-run default. A cloud profile becomes usable only when a valid model/provider selection and budget are present.

A status example should read: "All 421 source pages accounted for; 406 pages exported; 15 awaiting recognition; 38 formulas need review; 12 tables use temporary images; compilation passed for 9 of 10 chapters." These are illustrative numbers, not measured results. The dashboard must never compress them into a single green success indicator.

16. Revised implementation sequence

The attachments reduce design work and provide staged implementation contracts. They do not remove the converter, endpoint or fidelity gates. The sequence below replaces the earlier broad milestone list and keeps a runnable, reviewable result at each step.

Milestone	Deliverable and gate
0A. Reconcile inputs Plan alignment complete	Select standalone 1.1 and the 5 October skill; mark the older ZIP superseded. This revision records hashes, adaptation decisions and scope boundaries. Staging adapted project assets remains implementation work.
0B. Probe package Next implementation deliverable	Inspect any actual pilot code first. Adapt prompt 01 with two-Spark qualification: inventory, isolated converter trials, genuine GLM calls, vision result and paper/chapter measurements. No live probe is yet claimed.
1A. Source backbone	Adapt prompt 02. Register one paper, retain raw outputs/assets and source spans, prove complete page accounting, lock/crash recovery and backup/restore. Deliver exercised RUN-PILOT.md commands.
1B. One-paper LaTeX	Add typed blocks, native-renderer wrapper, GLM proposals, validation, compilation and review accept/edit/defer. Require resolving evidence, visible unresolved items and persistent human overrides.
2. Whole-book worker	Adapt prompt 05's book/release rules. Complete bounded jobs, chapter links, page ledger and full-book assembly; test interruption. Report conversion, editability and exact review depth separately.
3. Optional cloud assistance	Qualify image-capable model/provider profiles, bounded retries and budget/unknown-charge ledger. Compare mathematical errors and correction time. Capability probes may occur earlier if useful.
4. Interfaces and wiki bridge	Add browser convenience, optional Open WebUI adapter and complete-source export with reviewed ID mapping. Retrieval/digests/graph remain separate downstream work.

Concrete next package

Provide an adapted inventory/probe prompt, configuration template with unset machine-specific secrets, source-selection checklist, result manifest and launch/collection instructions. Discover the real endpoint and serving profile; do not copy historical addresses or model aliases as established facts. Keep the working TensorFold and system Python unchanged.

After those probes, implement ordinary Python stage functions and explicit state transitions. Reuse actual inspected components if present; otherwise implement the small source core from the contracts. Do not build a general autonomous agent framework or require a distributed queue.

Estimate book runtime only from representative measured page classes, model calls and build time. Estimate human correction effort separately. Exact dependency locks, memory coexistence, visual capability and native LaTeX limitations remain empirical questions.

Boundary: this update completes the revised plan for review. Prototype implementation, installations and live Spark runs remain the next stage under the agreed workflow. No installed skill, existing wiki or benchmark environment has been modified.

17. Sources and evidence (1/2)

Primary sources retrieved on 17 October 2026. Software pages can change; implementation must record exact versions and image/model hashes. Architecture, review policies and thresholds in this report are proposed design choices unless identified as upstream behaviour.

[1] MinerU release history
https://github.com/opendatalab/MinerU/releases
Candidate 4.0.12 and release timing; recheck at installation.

[2] MinerU quick start
https://opendatalab.github.io/MinerU/quick_start/
Python range, stateless CLI and page defaults.

[3] MinerU output formats and result contract
https://opendatalab.github.io/MinerU/reference/output_files/
LaTeX renderer versus CLI, Flash equation fallback and complete asset packages.

[4] MinerU tiers and runtimes
https://opendatalab.github.io/MinerU/usage/tiers/
Basic/Standard/Advanced roles and backend boundaries.

[5] MinerU NVIDIA deployment and Dockerfile
https://opendatalab.github.io/MinerU/quick_start/docker_deployment/
Runtime and pinning caveats. Architecture claim also checked in the official Dockerfile.

[6] NVIDIA DGX Spark known issues
https://docs.nvidia.com/dgx/dgx-spark/known-issues.html
Unified memory and memory-reporting limitations.

[7] Marker official repository
https://github.com/datalab-to/marker
Current modes, inline mathematics, OCR limits, outputs and runtime requirements.

[8] Docling enrichment documentation
https://docling-project.github.io/docling/usage/enrichments/
Formula recognition into LaTeX.

[9] Docling installation
https://docling-project.github.io/docling/getting_started/installation/
ARM64 support and component-specific constraints.

[10] Docling Serve
https://github.com/docling-project/docling-serve
Container deployment variants; qualify the exact ARM64/CUDA image.

[11] Chandra official repository
https://github.com/datalab-to/chandra
Optional page-level recognition challenger.

[12] LaTeX-OCR / pix2tex
https://github.com/lukas-blecher/LaTeX-OCR
Equation-image recognition, not a whole-document system.

18. Sources and evidence (2/2)

Primary sources retrieved on 17 October 2026. Software pages can change; implementation must record exact versions and image/model hashes. Architecture, review policies and thresholds in this report are proposed design choices unless identified as upstream behaviour.

[13] Nougat official repository
https://github.com/facebookresearch/nougat
Historical alternative and documented missing-page behaviour.

[14] PyMuPDF TextPage documentation
https://pymupdf.readthedocs.io/en/latest/textpage.html
Native text, image and coordinate evidence.

[15] Z.ai GLM-5.3-Flash overview
https://docs.z.ai/guides/vlm/glm-5.3-flash
Upstream multimodality; does not certify the local EXL3/TensorFold endpoint.

[16] OpenRouter image inputs
https://openrouter.ai/docs/guides/overview/multimodal/image-understanding
Explicit image requests and model/provider image limits.

[17] OpenRouter PDF inputs
https://openrouter.ai/docs/guides/overview/multimodal/pdfs
Native versus parsed PDF routing and OCR image limits.

[18] OpenRouter model catalogue API
https://openrouter.ai/docs/api/api-reference/models/list-all-models-and-their-properties
Discover model IDs, capabilities, limits and pricing.

[19] OpenRouter structured outputs
https://openrouter.ai/docs/guides/features/structured-outputs
Schema support must be qualified for the selected endpoint.

[20] OpenRouter provider routing
https://openrouter.ai/docs/guides/routing/provider-selection
Provider restrictions, fallback and unit-price controls.

[21] OpenRouter usage accounting
https://openrouter.ai/docs/cookbook/administration/usage-accounting
Returned usage/cost data; reconcile every attempt.

[22] Open WebUI tool development
https://docs.openwebui.com/features/extensibility/plugin/tools/development/
Current tool/configuration interfaces, not a version-0.12.0 compatibility certificate.

[23] Open WebUI v0.12.0 release
https://github.com/open-webui/open-webui/releases/tag/v0.12.0
The user's reported version exists; adapter testing is still required.

[24] latexmk package
https://ctan.org/pkg/latexmk
Build orchestration candidate for the pinned TeX environment.

Recovered project sources: Quant-Wiki-Practical-Plan.md 2.2; Quant-Wiki-Build-Contract.md 2.2; Quant Wiki Spark Pilot skill and its Walkthrough 1.1/Handover 2.2 provenance. The uploaded Guyon paper and Boyarchenko-Levendorskii book were sampled for pilot selection, not converted or benchmarked.

Revision 1.1 additions: attached Spark Pilot standalone walkthrough 1.1 (25 September), attached guide ZIP 1.0 (24 September), quant-wiki-spark-pilot skill revision 1.0 (5 October), and run-two-spark-models snapshot (4 October). Exact precedence, hashes and asset reuse are recorded in sections 3 and 3A. No new deployment evidence is claimed.